

Laboratory 3 Report Template:

Signals and Systems
CT/DT Systems and Basic System Properties
System Properties Explorer

Graded Submission Requirement

Deliverable (1/13): This is a graded assignment. Include your name and student ID in all required screenshots and in as many additional screenshots as possible. Screenshots without clear student identification may not receive full credit.

Deliverable (2/13): All MATLAB verification in this report must use your own captured hardware input/output data. Synthetic-only data is not accepted for verification credit.

Checkpoint (1/7): Before submission, verify every screenshot is legible and includes labels, units, and key measurements where applicable.

Student Information

Name 1	_____	Name 2	_____
NetID 1	_____	NetID 2	_____
Section	_____	Date	_____
TA	_____	Random Seed	_____

Note on Prompts

- **Pre-lab Deliverable (N):** Pre-lab work to complete before lab.
- **Checkpoint (N):** Items to show your TA in lab.
- **Discussion (N):** Analysis and interpretation in your report.
- **Deliverable (N):** Required report artifacts and evidence.

1 Assigned Parameters

Table 1. TA-Assigned Experiment Parameters

Parameter	Description	Assigned Value
f_1	Frequency for x_1	_____
f_2	Frequency for x_2	_____
a, b	Superposition coefficients	_____
Δ	Time shift for time-invariance test (ms)	_____
A_{max}	Maximum bounded-input amplitude	_____
Seed	Unique randomization seed	_____

2 Pre-Lab Results

Pre-lab Deliverable (1/1): Attach your complete pre-lab pages or include concise summaries below.

2.1 Predicted Property Truth Table

Table 2. Pre-Lab Property Predictions

System	Linear	TI	Causal	BIBO Stable	Pre-Lab Justification
RC filter					
Diode clipper					
Comparator					

2.2 Unique Test Signal Design Summary

- $x_1(t)$ definition: _____
- $x_2(t)$ definition: _____
- Superposition coefficients a, b : _____
- Time shift Δ and rationale: _____
- Bounded-input range used: _____

2.3 MATLAB Verification Plan (Summary)

Insert pseudocode or snapshot of your pre-lab MATLAB verification plan

3 Phase 1: Circuit Construction and Baseline Validation

3.1 Step 1.1 RC Filter Baseline

Checkpoint (2/7): Verify RC filter operation and record input/output relationship. **Deliverable (3/13):** Submit dual-channel scope screenshot with name/ID and annotations for amplitude ratio and delay.

Step 1.1 Evidence: RC filter scope screenshot

3.2 Step 1.2 Diode Clipper Baseline

Checkpoint (3/7): Verify clipping behavior and estimate clipping threshold from measurements. **Deliverable (4/13):** Submit scope screenshot with clipping threshold clearly labeled.

Step 1.2 Evidence: Diode clipper scope screenshot

3.3 Step 1.3 Comparator Baseline

Checkpoint (4/7): Verify threshold switching behavior of comparator output. **Deliverable (5/13):** Submit scope screenshot with analog input, comparator output, and threshold annotation.

Step 1.3 Evidence: Comparator scope screenshot

3.4 Circuit Photo Evidence

Deliverable (6/13): Submit one setup photo containing all three circuits (or three separate photos), each with student ID visible.

Phase 1 Evidence: Circuit photo(s) with visible student ID

4 Phase 2: Property Testing with Hardware Data

4.1 Step 2.1 Linearity (Superposition) Test

Checkpoint (5/7): Acquire data for x_1 , x_2 , and $ax_1 + bx_2$ for each circuit. **Deliverable (7/13):** Provide overlay plots and NRMSE for superposition agreement for all circuits.

Table 3. Linearity Verification Summary

Circuit	NRMSE (%)	Linear?	Evidence-Based Justification
RC filter	_____	_____	
Diode clipper	_____	_____	
Comparator	_____	_____	

Step 2.1 Evidence: MATLAB linearity overlays and residual plots

Discussion (1/5): Explain why each circuit passes or fails linearity using both circuit physics and numerical results.

4.2 Step 2.2 Time-Invariance Test

Checkpoint (6/7): Acquire data for $x(t)$ and $x(t - \Delta)$ for each circuit and compare outputs. **Deliverable (8/13):** Provide overlay plots of $\mathcal{T}\{x(t - \Delta)\}$ versus $y(t - \Delta)$ and report NRMSE.

Table 4. Time-Invariance Verification Summary

Circuit	NRMSE (%)	TI?	Evidence-Based Justification
RC filter	_____	_____	
Diode clipper	_____	_____	
Comparator	_____	_____	

Step 2.2 Evidence: MATLAB TI overlays and residual plots

Discussion (2/5): If mismatch occurs, separate true non-TI behavior from alignment, trigger, or sampling artifacts.

4.3 Step 2.3 Causality and Stability Checks

Checkpoint (7/7): Use bounded tests to verify no anticipatory output and bounded response. **Deliverable (9/13):** Submit scope evidence supporting causality and bounded-input bounded-output conclusions.

Table 5. Causality and Stability Summary

Circuit	Causal?	BIBO Stable?	Evidence-Based Justification
RC filter			
Diode clipper			
Comparator			

Step 2.3 Evidence: Scope captures for causality and bounded-output tests

5 Phase 3: MATLAB Verification from Captured Data

5.1 Step 3.1 Data Import and Alignment

Deliverable (10/13): Submit MATLAB code showing import, cleaning, and alignment of acquired datasets.

Step 3.1 Evidence: MATLAB import/alignment code screenshot

5.2 Step 3.2 Numerical Metrics Overview

Deliverable (11/13): Report final NRMSE values used to support property conclusions.

Table 6. Numerical Verification Summary

Circuit	Linearity NRMSE (%)	TI NRMSE (%)
RC filter	_____	_____
Diode clipper	_____	_____
Comparator	_____	_____

5.3 Step 3.3 Final Property Truth Table

Deliverable (12/13): Complete the final truth table based on all scope and MATLAB evidence.

Table 7. Final Property Conclusions

System	Linear	TI	Causal	BIBO Stable	One-Sentence Final Justification
RC filter					
Diode clipper					
Comparator					

Deliverable (13/13): Attach a handwritten property truth table image with your ID visible.

Step 3.3 Evidence: Handwritten truth table with student ID

6 Discussion and Interpretation

Discussion (3/5): For each circuit, identify the physical mechanism that most strongly determines its property outcomes (memory, thresholding, clipping, switching).

Discussion (4/5): Give one concrete example from your data showing how an LTI approximation can be useful but incomplete for real hardware.

Discussion (5/5): Describe one failure mode in data collection (synchronization, trigger, quantization, or noise) that can mislead property verification if not controlled.

Final Submission Checklist

- ☐ Pre-lab pages and prediction table included.
- ☐ Circuit photo evidence with visible student ID.
- ☐ RC, clipper, and comparator scope screenshots included and annotated.
- ☐ MATLAB superposition verification plots and NRMSE values included.
- ☐ MATLAB time-invariance verification plots and NRMSE values included.
- ☐ Causality and BIBO evidence included for all circuits.
- ☐ Final property truth table completed and justified.
- ☐ MATLAB scripts shown to use captured hardware data.